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General Comment

To the Office of Science and Technology Policy,

AI & Partners applauds the White House's commitment to fostering AI innovation while ensuring security, ethical governance, and economic growth. We urge policymakers to prioritize public-private partnerships, robust regulatory frameworks, and sustainable AI infrastructure. Transparency, fairness, and security must be foundational to AI development, particularly in high-risk applications. By balancing innovation with responsible oversight, the U.S. can lead global AI governance while safeguarding national interests. We appreciate the opportunity to contribute and look forward to policies that support trustworthy AI for all.

Thank you,

Michael Charles Borrelli

AI & Partners, B.V.

Attachments

OSTP - AI Action Plan- Request for Information - 2025-03-12

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The White House

Office of Science and Technology Policy (OSTP)

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Request for Information on Artificial Intelligence Action Plan[1]

To the Office of Science and Technology Policy,

AI & Partners appreciates the opportunity to contribute to the ongoing efforts of the U.S. government in shaping the future of artificial intelligence (AI). As a global leader in AI governance, we commend the White House's commitment to ensuring that AI development aligns with American values of innovation, security, and economic competitiveness. This letter outlines our recommendations for advancing trustworthy AI innovation, establishing global leadership in AI governance, and ensuring AI remains a force for economic growth while mitigating security risks.

The advancement of AI is pivotal to ensuring the United States remains the global leader in innovation, economic growth, and national security. In response to Executive Order 14179[2], this document outlines key policy recommendations for the AI Action Plan, focusing on regulatory frameworks that foster innovation while maintaining oversight and eliminating unnecessary barriers that hinder AI development and deployment.

A well-balanced AI policy should ensure responsible innovation, prevent monopolistic control, and uphold ethical standards while encouraging rapid development. Given AI's pervasive influence across industries, the U.S. must implement a legal and regulatory structure that supports growth without stifling competition or risking security. This document aims to highlight necessary legislative, regulatory, and governance measures to secure America's AI leadership.

1. Push Multi-Stakeholder Collaboration Within Regulatory Confines

1.1 Foster Public-Private Partnerships and Compliance Frameworks

Collaboration between government agencies, academia, and the private sector is essential for ensuring AI hardware development aligns with national security and innovation policies.

Regulatory bodies should develop compliance frameworks that incentivize AI chip innovation while addressing intellectual property concerns, antitrust issues, and export control laws. Research grants should be structured with legal requirements that encourage compliance with national interests.

[1] https://www.federalregister.gov/documents/2025/02/06/2025-02305/request-for-information-on-the-development-of-an-artificial-intelligence-ai-action-plan
[2] https://www.federalregister.gov/documents/2025/01/31/2025-02172/removing-barriers-to-american-leadership-in-artificial-intelligence

Strategic Policy Recommendations:

- Public-Private Research Initiatives: The government should establish collaborative research programs with private semiconductor firms and academic institutions to accelerate AI hardware innovation. These initiatives should focus on developing energy-efficient AI chips, quantum computing capabilities, and next-generation semiconductor materials.
- Export Control Laws: Regulatory agencies should refine and enforce export control laws to prevent sensitive AI hardware technologies from being acquired by adversarial nations. Clear guidelines should be set regarding the export of AI chips and related intellectual property to maintain a competitive advantage in AI development.
- Compliance and Security Standards: The establishment of clear security standards for AI chip manufacturing will prevent vulnerabilities and ensure that AI hardware meets national security requirements. Government agencies should work closely with industry leaders to enforce compliance with cybersecurity best practices.
- Investment in Workforce Development: Training and education programs should be expanded to equip the workforce with the necessary skills to support AI chip manufacturing. Universities and technical colleges should collaborate with AI hardware companies to create specialized curricula that address industry demands.

Why this matters?

A robust compliance framework ensures that investments in AI hardware do not lead to monopolization or intellectual property theft. The dominance of a few corporations in the semiconductor industry could stifle innovation and limit competition, making it essential to implement regulations that encourage fair market practices. Additionally, strong intellectual property protections will prevent foreign competitors from illicitly acquiring U.S.-developed AI chip technologies.

Public-private partnerships should operate within a legal structure that safeguards U.S. technological interests, preventing adversarial nations from gaining access to critical AI infrastructure. In recent years, concerns over foreign acquisitions of semiconductor firms have highlighted the need for stringent investment review mechanisms. Strengthening oversight in AI hardware investments will ensure that U.S. advancements remain secure and competitive.

Moreover, AI chip manufacturing requires a highly skilled workforce. Addressing talent shortages through education and training programs will ensure that the U.S. remains at the forefront of semiconductor innovation. Government funding for AI hardware research should be accompanied by workforce development initiatives to create a pipeline of engineers, scientists, and technicians equipped to support the evolving demands of the industry.

2. Encourage Data Center Development While Driving Energy Efficiency

As AI adoption continues to expand, the infrastructure supporting AI models—including data centers and cloud computing systems—must evolve to meet increasing computational demands. The rapid proliferation of AI applications has led to significant energy consumption, placing strain on power grids and raising concerns about sustainability. Ensuring that AI infrastructure operates efficiently while maintaining security and environmental responsibility is paramount to achieving long-term technological and economic stability.

2.1 Drive Regulatory Frameworks for Sustainable AI Infrastructure

Given AI's intensive energy consumption, policymakers must implement regulations that mandate energy-efficient data center operations. Legislative measures should require AI-driven companies to meet energy efficiency standards, invest in renewable energy, and follow sustainability reporting guidelines. Compliance incentives such as tax breaks should be offered to companies that adopt green computing practices.

Strategic Policy Recommendations:

- Mandatory Energy Efficiency Standards: The government should establish regulations requiring data centers to meet stringent energy efficiency benchmarks. This includes standards for power usage effectiveness (PUE), cooling efficiency, and overall carbon footprint reduction.
- Renewable Energy Investments: Companies operating large-scale AI data centers should be incentivized to transition to renewable energy sources, such as solar, wind, and hydroelectric power, through tax credits and subsidies.
- Sustainability Reporting Requirements: AI companies should be required to disclose their energy consumption, carbon emissions, and sustainability initiatives in regular reports. Transparent reporting will encourage accountability and industry-wide adoption of green practices.
- Incentivizing Innovation in Energy-Efficient AI Hardware: Investment in energy-efficient AI chips and cooling technologies should be prioritized. Research and development grants can be allocated to companies pioneering advancements in low-power AI processing.
- Smart Grid Integration: AI data centers should be integrated with smart grid technologies that optimize energy distribution and reduce peak-load demand. This will enhance grid resilience and lower the overall energy impact of AI operations.

Rationale:

AI processing consumes vast amounts of energy, and inefficient data centers contribute significantly to carbon emissions. Without sustainable policies, the expansion of AI-driven applications could exacerbate climate change and strain national energy resources. Implementing regulatory measures to improve energy efficiency will ensure AI development does not come at the cost of environmental degradation, aligning AI growth with national climate goals.

The demand for computational power is expected to surge with advancements in deep learning and large-scale AI models. To mitigate environmental consequences, a multi-faceted approach combining regulatory oversight, private sector initiatives, and technological innovation is necessary. Encouraging the adoption of energy-efficient hardware and promoting clean energy solutions will help balance AI's transformative potential with environmental sustainability.

2.2 Encourage Policies on Cloud and Edge Computing Expansion

Expanding edge computing capabilities requires regulatory safeguards to ensure data security and minimize cyber threats. Policymakers should establish national guidelines for decentralized AI data

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processing, ensuring that edge AI development adheres to data protection laws. Cybersecurity regulations should be enhanced to mitigate vulnerabilities associated with distributed AI systems.

Strategic Policy Recommendations:

- National Standards for Edge Computing Security: Regulatory agencies should establish uniform security standards for decentralized AI processing. These standards should include encryption requirements, data anonymization protocols, and compliance with existing data protection laws.
- Stronger Cybersecurity Regulations: Edge computing introduces unique security risks due to its decentralized nature. New policies should mandate AI companies to implement robust security frameworks, including end-to-end encryption, multi-factor authentication, and AI-driven threat detection systems.
- Data Sovereignty and Compliance: AI applications that rely on cloud and edge computing should comply with national data sovereignty laws. Companies should be required to process and store sensitive data within national borders to prevent unauthorized foreign access.
- Investment in Secure Edge Infrastructure: The government should provide funding for research and development in secure edge computing technologies. This includes innovations in hardware security modules (HSMs), secure processing units (SPUs), and tamper-resistant AI chips.
- Collaboration with Industry Stakeholders: Public-private partnerships should be established to create best practices for secure cloud and edge AI deployments. Industry leaders, government agencies, and cybersecurity experts should collaborate on developing threat mitigation strategies.

Rationale:

As AI moves toward decentralized processing models, ensuring robust security in cloud and edge computing is essential. Traditional cloud computing models concentrate data in centralized servers, making them prime targets for cyberattacks. The shift toward edge AI—where processing occurs on local devices rather than centralized data centers—reduces latency and enhances efficiency but introduces new security challenges.

Without stringent regulations, decentralized AI systems could become vulnerable to data breaches, unauthorized access, and cyber threats. Government regulations should mandate best practices to prevent security breaches, protect consumer data, and maintain system integrity across decentralized AI applications. Furthermore, clear guidelines on data storage and processing locations will enhance compliance with privacy laws and safeguard against foreign cyber espionage.

As a result of prioritizing security in cloud and edge computing, policymakers can support the growth of AI while ensuring data integrity and user privacy. The future of AI infrastructure must be built on a foundation of security, efficiency, and sustainability to unlock the full potential of AI without compromising national and global interests.

3. Reinforce Trustworthy AI Model Development and Open-Source AI

As AI systems continue to evolve, ensuring ethical, transparent, and secure development practices is essential. AI models influence decision-making in critical sectors, including healthcare, finance, and law enforcement, making regulatory oversight a necessity. At the same time, open-source AI presents opportunities for innovation but also raises concerns regarding security and misuse. Policymakers must strike a balance between fostering AI advancement and mitigating risks through comprehensive legal and regulatory frameworks.

3.1 Prioritise Ethical and Transparent AI Models

Regulatory frameworks should require AI models, especially those used in critical sectors, to adhere to transparency and explainability mandates. Government oversight should ensure that AI algorithms used in healthcare, finance, and law enforcement comply with non-discrimination laws and fairness standards. Regular audits should be mandated to assess AI systems for biases and ethical concerns.

Strategic Policy Recommendations:

- Mandatory Transparency and Explainability Standards: AI models should be designed to provide clear and interpretable explanations for their decisions. Developers must document and disclose model behavior, decision-making criteria, and data sources.
- Bias Auditing and Fairness Compliance: AI systems should undergo regular independent audits to identify and mitigate biases, ensuring compliance with anti-discrimination laws. Companies deploying AI in critical areas must submit fairness reports outlining potential biases and corrective measures.
- Sector-Specific AI Regulations: AI models used in healthcare, finance, and law enforcement should be subject to sector-specific guidelines. For instance, healthcare AI must comply with patient privacy laws, while financial AI should align with fair lending practices.
- AI Governance Boards and Ethical Review Committees: Establishing government-backed AI oversight committees can ensure that high-risk AI applications adhere to ethical standards. These committees should be composed of legal experts, ethicists, technologists, and industry representatives.
- Public Disclosure for High-Impact AI Systems: AI models influencing public policy, legal decisions, or essential services should be required to publish transparency reports detailing their algorithms, limitations, and ethical considerations.

Rationale:

AI models have the potential to reinforce systemic biases if not properly regulated. Unchecked AI decision-making can lead to discrimination in hiring, lending, medical diagnosis, and law enforcement. Legal requirements for transparency will help ensure AI-driven decisions are fair, accountable, and justifiable, preventing discrimination and ethical violations.

The need for fairness in AI systems is evident from past incidents where biased AI models disproportionately affected marginalized groups. Implementing strict auditing and explainability standards will increase public trust in AI applications while promoting responsible innovation. Ensuring that AI models remain interpretable and explainable will also facilitate regulatory compliance and reduce the risks associated with opaque algorithms making critical decisions.

3.2 Ensure Legal Protections and Licensing for Open-Source AI

Policymakers should implement legal safeguards for open-source AI development to balance innovation with security risks. Intellectual property laws must be adapted to address AI-generated content and licensing agreements. National AI repositories should be governed by legal frameworks that promote responsible use while preventing malicious exploitation.

Strategic Policy Recommendations:

- Licensing and Usage Agreements: Open-source AI projects should be required to include legally binding licensing agreements specifying ethical usage guidelines. These agreements should prohibit the development of AI for malicious purposes, such as deepfake creation, automated disinformation campaigns, and cyberattacks.
- Intellectual Property Adaptations for AI-Generated Content: Current IP laws must be updated to address AI-generated content, including determining ownership rights for AI-created works and protecting open-source contributors from unauthorized commercial exploitation.
- National AI Repositories: Establishing government-regulated AI repositories can ensure responsible access to open-source AI tools. These repositories should vet AI models for ethical compliance before allowing public distribution.
- Risk Assessment Frameworks for Open-Source Contributions: Developers should be required to perform security and ethical risk assessments when contributing to open-source AI projects. Regulatory bodies can provide guidelines to classify AI models based on their potential risks and societal impact.
- International Collaboration on Open-Source AI Governance: AI policy should be harmonized across international jurisdictions to prevent regulatory gaps that allow malicious actors to exploit open-source AI tools. A global AI ethics consortium can facilitate collaboration among governments, research institutions, and private-sector stakeholders.

Rationale:

Open-source AI facilitates innovation but also introduces risks, including misuse by bad actors. While open collaboration accelerates AI research and democratizes access to cutting-edge technology, it also raises security concerns, particularly when AI tools can be repurposed for harmful applications.

For instance, AI-generated deepfakes have been used for political misinformation, fraud, and identity theft. Without proper safeguards, open-source AI could also be leveraged for cyberattacks, automated hacking tools, or unethical surveillance practices. Licensing frameworks will help maintain control over AI development while allowing open collaboration in research and development.

3.3 Support Private Sector AI Regulation

Policies should foster AI innovation in the private sector while enforcing consumer protection laws. Regulatory frameworks must address AI's impact on labor markets, consumer data rights, and ethical AI use in corporate environments. Standards for AI liability should be established to ensure accountability for AI-driven decisions that impact consumers.

Strategic Policy Recommendations:

- Consumer Protection and AI Transparency: Companies using AI-driven products and services should be required to disclose how AI models impact consumers, particularly in critical sectors such as finance, healthcare, and employment.
- Ethical AI Development Standards: Corporations should implement ethical guidelines to prevent AI misuse, including bias mitigation strategies and compliance with responsible AI frameworks.
- AI Impact Assessments: Businesses should conduct AI impact assessments before deploying AI technologies that could significantly affect consumer rights, employment, or public safety. These assessments should be reviewed by regulatory bodies to ensure compliance.
- AI and Labor Market Protections: Policies should address the impact of automation on employment, providing workforce transition programs, retraining initiatives, and AI-related job creation incentives.
- Data Privacy and AI Liability Laws: Strengthening AI-specific data privacy laws will help protect consumer information. Companies should also be held accountable for AI-generated decisions that result in harm, ensuring legal recourse for affected consumers.
- Competition and Antitrust Measures: Regulations should prevent monopolistic practices in AI markets, ensuring fair competition and preventing large corporations from stifling innovation through restrictive AI patents and proprietary models.

Rationale:

Unchecked AI development in the private sector may lead to unethical practices, monopolization, or labor displacement. AI-driven decision-making can affect individuals' access to jobs, financial services, and healthcare, making regulatory oversight essential. For example, AI-powered hiring tools must be scrutinized to prevent biased employment practices, and AI-based loan approval systems must comply with anti-discrimination laws.

4. Require Explainability and Assurance of AI Models

As AI systems become more integrated into critical aspects of society, ensuring their transparency, fairness, and reliability is essential. High-risk AI applications, such as those used in healthcare, finance, law enforcement, and autonomous systems, must be subject to rigorous oversight. Explainability and assurance frameworks will help build public trust in AI-driven technologies by ensuring that AI decisions are interpretable, ethical, and compliant with legal standards.

4.1 Foster Transparent AI Decision-Making

AI models used in high-risk applications should be subject to mandatory explainability requirements. Regulations should require AI systems in autonomous vehicles, healthcare, and financial decision-making to provide interpretable outputs. AI assurance frameworks should standardize evaluation methodologies for algorithmic transparency.

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Strategic Policy Recommendations:

- Mandatory Explainability Standards: AI developers should be required to design models that provide clear explanations for their decisions. This is particularly crucial in fields such as medical diagnostics, loan approvals, and automated legal assessments, where opaque decision-making could result in significant consequences.
- Sector-Specific Explainability Guidelines: Different industries require different levels of AI interpretability. For example, in healthcare, AI diagnostic models should be required to generate justifications for their recommendations, while in finance, automated lending algorithms must demonstrate fairness and risk assessment transparency.
- Standardized AI Transparency Frameworks: AI assurance methodologies should be established to assess the explainability of models. These frameworks should define measurable transparency criteria and create benchmarks for compliance.
- User-Friendly AI Explanations: AI systems interacting with consumers should provide understandable explanations for their decisions. For example, AI-powered hiring tools should offer applicants insights into why they were or were not selected.
- Explainability Testing in AI Certification: Before deployment, AI systems should undergo rigorous explainability assessments to ensure compliance with legal and ethical requirements. These tests should be conducted by independent auditors and regulatory bodies.

Rationale:

Opaque AI decision-making can lead to distrust and systemic bias. AI systems that operate as "black boxes" create risks by making decisions without human interpretability, making it difficult to challenge or correct errors. In high-stakes fields, such as medical treatment planning or criminal justice, a lack of transparency can result in serious harm to individuals and erode public trust in AI technologies.

Mandatory explainability requirements will ensure AI-driven decisions are ethical, legally compliant, and understandable to both experts and consumers. Transparency frameworks will also encourage AI developers to adopt best practices in ethical AI design, mitigating risks associated with biased or discriminatory decision-making.

4.2 Implement AI Auditing and Compliance Regulations

An AI regulatory body should oversee compliance auditing to ensure AI models meet legal, ethical, and safety standards. Certification requirements should be established for AI-driven products to ensure adherence to regulatory frameworks before deployment.

Strategic Policy Recommendations:

- Creation of AI Oversight Agencies: Governments should establish independent regulatory bodies responsible for auditing AI models, ensuring they meet transparency, fairness, and safety standards before deployment.
- Mandatory AI Compliance Audits: AI-driven systems should be subject to regular compliance audits conducted by independent reviewers. These audits should assess bias, ethical concerns, and security vulnerabilities in AI algorithms.

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- **Certification for AI Deployment**: AI systems used in high-risk applications should obtain regulatory certification before public use. Certification criteria should include explainability standards, bias assessments, and security evaluations.
- **AI Model Documentation Requirements**: AI developers should be required to maintain detailed documentation on model training data, algorithmic changes, and decision-making processes. This documentation should be accessible to auditors and regulatory agencies.
- **Whistleblower Protections for AI Ethics Violations**: Employees and researchers who expose unethical AI practices should be granted legal protections, ensuring they can report AI-related misconduct without fear of retaliation.

Rationale:

AI audits help prevent errors, discrimination, and security vulnerabilities. Without proper oversight, AI models can reinforce biases, make erroneous decisions, or become targets for adversarial attacks. Auditing and certification processes will ensure that AI systems adhere to ethical and legal standards before they impact consumers and businesses.

A standardized compliance framework will also provide clarity for AI developers, setting clear expectations for responsible AI deployment. Establishing independent regulatory agencies will create a mechanism for continuous AI oversight, ensuring long-term accountability and consumer protection.

Thank you for considering our comments. We look forward to seeing the final Artificial Intelligence Action Plan in place.

Kind regards,

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